

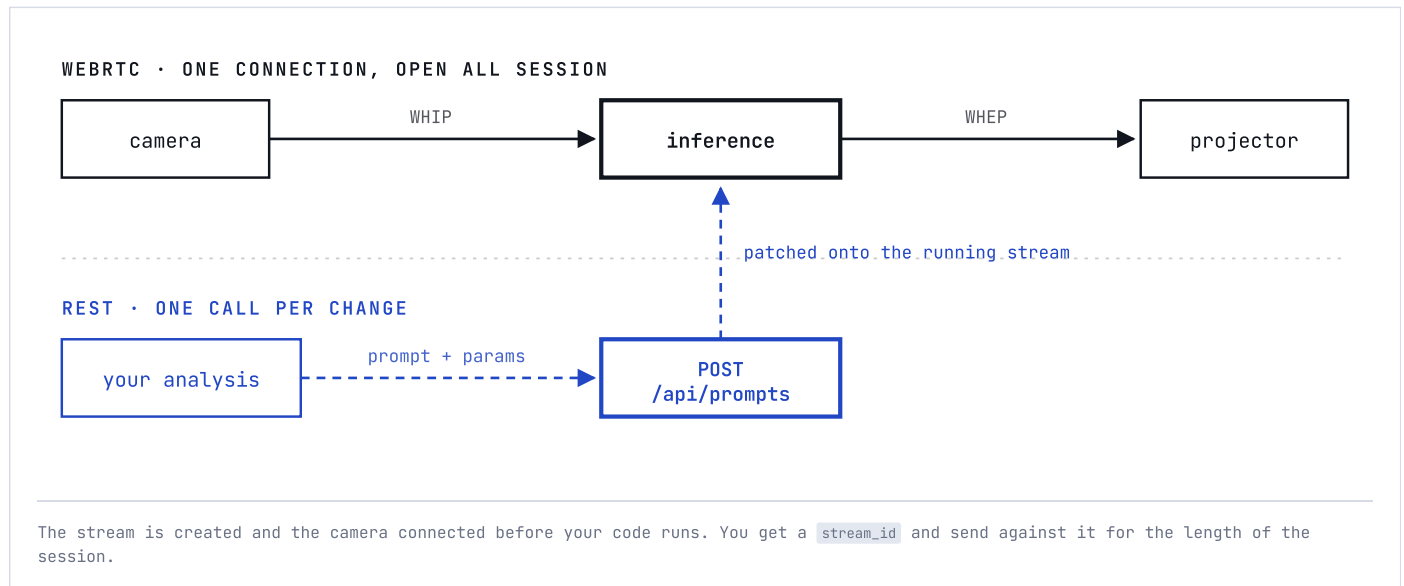
The seed42 Control API

Request shape, settable fields and their ranges, which fields update on a running pipeline and which reload it, and the client's timing behaviour. SDXL only.

TRANSPORT

Persistent for video, REST for control

Video runs over one WebRTC connection that stays open for the session, WHIP in and WHEP out. Parameters are separate: a REST call each time something changes, patched onto the stream while it runs. The stream is never re-created for a parameter change.



THE REQUEST

Partial updates merge

One endpoint. Whatever is in `body` is forwarded to the engine as that stream's parameters. A **partial object merges** with the existing configuration rather than replacing it; fields not present are left at their current values.

```
POST /api/prompts
Content-Type: application/json

{
  "stream_id": "<the running stream>",
  "body": {
    "prompt": "concrete hallway, hard shadows, deep red",
    "t_index_list": [ <int> ],
    "controlnets": [
      { "conditioning_scale": <0-1> }, // depth
      { "conditioning_scale": <0-1> }, // canny
      { "conditioning_scale": <0-1> }  // tile
    ]
  }
}
```

Upstream this becomes `PATCH /v1/streams/{id}` with `{ pipeline: "streamdiffusion", params: { ... } }`. The engine key is held by seed42 and is not present in the request.

STATUS	MEANING
200	Applied to the running stream
400	<code>stream_id</code> or <code>body</code> missing
401	No valid seed42 session on the request
404 + "not ready"	Stream exists but has not finished starting
other	Passed through from the engine unchanged, with its <code>error</code> , <code>code</code> and <code>requestId</code>

THE RULE THAT MATTERS MOST

Hot-swap, or a 30 second reload

A documented subset of fields updates on the running pipeline. Every other field triggers a pipeline reload, documented at approximately thirty seconds, during which the stream produces no output.

FIELD	BEHAVIOUR	NOTES
prompt	HOT	String, or weighted tuples. See below
negative_prompt	HOT	String
t_index_list	HOT	1 to 4 integers, non-decreasing
guidance_scale	HOT	Number
delta	HOT	Number
num_inference_steps	HOT	Integer, 1 to 100. Also affects output frame rate
seed	HOT	Integer, or weighted tuples
controlnets[.conditioning_scale]	HOT	0 to 1. The scale only. Changing a ControlNet's model or preprocessor is not hot
ip_adapter.*	HOT	Scale, enabled, weight type, and the style image URL
model_id	RELOAD	Set at stream creation
width / height	RELOAD	Integer, 384 to 1024, divisible by 64. Set at stream creation
controlnets[.model_id / preprocessor]	RELOAD	Omitted from a partial update, they are left unchanged by the merge
acceleration, lora, filters	RELOAD	Set at stream creation

TYPES

Weighted prompts and seeds

`prompt` accepts either a plain string or an array of `[string, number]` tuples. `seed` accepts either an integer or an array of `[integer, number]` tuples. Both fields are hot-swappable in either form.

```
"prompt": [
  ["concrete hallway, hard shadows", 0.7],
  ["deep red haze, slow drift", 0.3]
]
```

Two settings govern how weighted values are combined, both set at stream creation: `prompt_interpolation_method` (`linear` or `slerp`) and `normalize_prompt_weights` . With normalisation enabled the weights are relative and need not sum to 1. The seed equivalents are `seed_interpolation_method` and `normalize_seed_weights` .

MAPPING

Our labels, and the fields behind them

OUR LABEL	FIELD	WHAT IT IS
Prompt	prompt	String or weighted tuples
Negative	negative_prompt	String
Creativity	t_index_list	Denoising start indices. 1 to 4 integers, non-decreasing; we send one. Lower values diverge further from the camera, higher stay closer. Our UI shows it inverted, so the number a user sees is not the number sent
Depth	controlnets[0].conditioning_scale	xinsir/controlnet-depth-sdxl-1.0 , depth_tensorrt preprocessor
Edge	controlnets[1].conditioning_scale	xinsir/controlnet-canny-sdxl-1.0 , canny preprocessor
Consistency	controlnets[2].conditioning_scale	xinsir/controlnet-tile-sdxl-1.0 , feedback preprocessor. The previous output fed back, which is what holds frames steady against each other
Image ref	ip_adapter_style_image_url	HTTPS URL or base64 data URI, 5MB max
Image ref strength	ip_adapter.scale	Only read when ip_adapter.enabled is true
Steps	num_inference_steps	1 to 100
Seed	seed	Integer or weighted tuples

The three ControlNets are **positional** and fixed in that order for the session. In a partial update, entries containing only `conditioning_scale` leave the remaining keys of each object unchanged.

The image reference, in more detail

- Accepted forms: an `https` URL, or a base64 `data:` URI. The documented ceiling is **5MB**.
- The value travels inline in the request body on every call that includes it. An `https` URL sends only the URL, and must be reachable by the engine.
- The seed42 client downscales references to a 768px long edge and re-encodes as JPEG before sending.
- `ip_adapter` also takes `type` (`regular` or `faceid`) and `weight_type`, an enum of named blending curves including `style transfer` and `composition`. Both are left at their defaults.

FIXED ON OUR SIDE

Set once, and not yours to change

Set when the stream is created and unchanged for the session. Listed for completeness; their values are not exposed.

FIELD	WHY IT IS FIXED
model_id	stabilityai/sdxl-turbo. The only model we run
width, height	Changing them reloads the pipeline
controlnets[].model_id, preprocessor, preprocessor_params, enabled, control_guidance_start, control_guidance_end	The three-ControlNet stack is the recipe. Only the scales move
prompt_interpolation_method, normalize_prompt_weights, seed_interpolation_method, normalize_seed_weights	How blending behaves. Tuned once
acceleration, use_lcm_lora, lcm_lora_id, lora_dict, do_add_noise, use_denoising_batch	Engine setup. Reload on change
enable_similar_image_filter and its thresholds, skip_diffusion	Frame-skipping behaviour we have settled on
cached_attention, image and latent pre/post-processing	Available in the engine, not part of our recipe

TIMING

Rate limits and timing behaviour

- seed42 applies no rate limit to this endpoint. The engine has its own limits, which the seed42 client has not encountered and does not handle. The behaviour below is what the seed42 client implements.
- **One request in flight per stream.** A change arriving while a call is open replaces the pending payload rather than queueing behind it. Last write wins.
 - **Requests are aborted after 12 seconds.**
 - **Retries.** A 404 whose body contains "not ready" is retried up to 3 times, 2 seconds apart. No other status is retried.
 - **Ordering.** Parameters are sent only after the WHIP connection is established. Calls before that return "not ready".
 - **Application.** A change applies to frames produced after it arrives, not to frames already in flight.

MODELS

What is running on our side

SDXL-Turbo (stabilityai/sdxl-turbo) on a StreamDiffusion pipeline. It is the only model in use, and model_id is fixed to it for the session.

IP-Adapter and cached attention are available on this model. Serving and acceleration are handled by the inference provider.

Field names, types, ranges, the hot-swap list and the merge behaviour are from the Daydream create-stream and update-stream references, checked 20 Aug 2026. Everything about what seed42 fixes, and the client timing behaviour, is from the seed42 implementation.